

Eigenvalues and eigenvectors

Basic skills

Q1. a) $\begin{vmatrix} x & 5 \\ 2 & x \end{vmatrix} = x^2 - 10$ so for singular
 $x^2 - 10 = 0 \Rightarrow x = \pm\sqrt{10}$

b) $\begin{vmatrix} -1+x & -3 \\ 2-x & 5 \end{vmatrix} = -5 + 5x - (-6 + 3x) = 1 + 2x$
for singular $1 + 2x = 0 \Rightarrow x = -\frac{1}{2}$

Q2. $M - \lambda I_3 = \begin{pmatrix} 4-\lambda & 2 & -3 \\ 5 & 9-\lambda & 2 \\ 0 & -1 & 5-\lambda \end{pmatrix}$

so $|M - \lambda I_3| = (4-\lambda) \begin{vmatrix} 9-\lambda & 2 \\ -1 & 5-\lambda \end{vmatrix} - 2 \begin{vmatrix} 5 & 2 \\ 0 & 5-\lambda \end{vmatrix} - 3 \begin{vmatrix} 5 & 9-\lambda \\ 0 & -1 \end{vmatrix}$

$$= (4-\lambda) ((9-\lambda)(5-\lambda) + 2) - 2(5(5-\lambda)) - 3(-5)$$
$$= (4-\lambda)(9-\lambda)(5-\lambda) + 2(4-\lambda) - 10(5-\lambda) + 15$$

$$= (4-\lambda)(45-14\lambda+\lambda^2) + 8-2\lambda - 50+10\lambda+15$$

$$= -\lambda^3 + 18\lambda^2 - 101\lambda + 180 - 27 - 8\lambda$$

$$= -\lambda^3 + 18\lambda^2 - 109\lambda + 153$$

Q3. $M = \begin{pmatrix} 3 & 5 \\ 0 & -4 \end{pmatrix}$

a) $M - \lambda I = \begin{pmatrix} 3-\lambda & 5 \\ 0 & -4-\lambda \end{pmatrix}$

$$|M - \lambda I| = (3-\lambda)(-4-\lambda) = -(3-\lambda)(4+\lambda)$$

b) For $|M - \lambda I| = 0 \Rightarrow \lambda = 3$ or -4

c) $M \underline{v} = 3 \underline{v}$ so $\begin{pmatrix} 3 & 5 \\ 0 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix}$

$$\Rightarrow 3x + 5y = 3x, \quad -4y = 3y$$

so $y = 0$ any x so $\begin{pmatrix} x \\ 0 \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

or $(M - 3I) \underline{v} = \underline{0} \Rightarrow \begin{pmatrix} 0 & 5 \\ 0 & -7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$
 $\Rightarrow y = 0$

Competency skills

Q4. $M \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$ so $\begin{pmatrix} -8 & -4 \\ 25 & 12 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix}$

$$\Rightarrow \begin{aligned} -8x - 4y &= 2x \\ 25x + 12y &= 2y \end{aligned} \Rightarrow \begin{aligned} 4y &= 10x \\ 10x &= -25x \end{aligned}$$

so $y = \frac{5}{2}x$ so $\begin{pmatrix} x \\ \frac{5}{2}x \end{pmatrix}$ e.g. $\begin{pmatrix} 2 \\ 5 \end{pmatrix}$

Q5. a) $M = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$

$$|M - \lambda I| = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 0 \\ 0 & 5-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)(5-\lambda) = 0 \text{ so } \lambda = 2, 5$$

For $\lambda = 2$:

$$\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 2x &= 2x \\ 5y &= 2y \end{aligned}$$

so $y = 0$ any x so $\begin{pmatrix} x \\ 0 \end{pmatrix}$ e.g. $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

For $\lambda = 5$:

$$\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 5 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 2x &= 5x \\ 5y &= 5y \end{aligned}$$

so $x = 0$ any y so $\begin{pmatrix} 0 \\ y \end{pmatrix}$ e.g. $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

b) $\begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix}$

$$|M - \lambda I| = 0 \Rightarrow \begin{vmatrix} 3-\lambda & 1 \\ 5 & -1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(-1-\lambda) - 5 = 0 \Rightarrow \lambda^2 - 2\lambda - 3 - 5 = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 8 = 0 \Rightarrow (\lambda - 4)(\lambda + 2) = 0$$

so $\lambda = -2$ or 4

For $\lambda = -2$:

$$\begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = -2 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 3x + y &= -2x \\ 5x - y &= -2y \end{aligned}$$

$$\Rightarrow y = -5x \quad \text{so} \quad \begin{pmatrix} x \\ -5x \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ -5 \end{pmatrix}$$

For $\lambda = 4$:

$$\begin{pmatrix} 3 & 1 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 4 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 3x + y &= 4x \\ 5x - y &= 4y \end{aligned}$$

$$\Rightarrow y = x \quad \text{so} \quad \begin{pmatrix} x \\ x \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

c) $M = \begin{pmatrix} 5 & 1 \\ 4 & 5 \end{pmatrix}$ so $|M - \lambda I| = 0 \Rightarrow (5-\lambda)^2 - 4 = 0$

$$\Rightarrow \lambda^2 - 10\lambda + 25 - 4 = 0 \Rightarrow \lambda^2 - 10\lambda + 21 = 0$$

$$\text{so } (\lambda - 3)(\lambda - 7) = 0 \Rightarrow \lambda = 3, 7$$

For $\lambda = 3$:

$$\begin{pmatrix} 5 & 1 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 3 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 5x + y &= 3x \\ 4x + 5y &= 3y \end{aligned}$$

$$\Rightarrow y = -2x \quad \text{so } \begin{pmatrix} x \\ -2x \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

For $\lambda = 7$:

$$\begin{pmatrix} 5 & 1 \\ 4 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 7 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow \begin{aligned} 5x + y &= 7x \\ 4x + 5y &= 7y \end{aligned}$$

$$\Rightarrow y = 2x \quad \text{so } \begin{pmatrix} x \\ 2x \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Q6.

$$M = \begin{pmatrix} 2 & 5 & 3 \\ 0 & 9 & -1 \\ 0 & 0 & -5 \end{pmatrix}$$

As upper triangular $|M - \lambda I| = (2 - \lambda)(9 - \lambda)(-5 - \lambda)$

so for eigenvalues $-(2 - \lambda)(9 - \lambda)(5 + \lambda) = 0$

$$\Rightarrow \lambda = -5, 2, 9$$

For $\lambda = -5$:

use $(M - \lambda I)v = \underline{0}$

$$\begin{pmatrix} 7 & 5 & 3 \\ 0 & 14 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{So } 7x + 5y + 3z = 0$$

$$14y - z = 0$$

$$\Rightarrow z = 14y \Rightarrow 7x + 5y + 42y = 0$$

$$\text{So } x = -\frac{47}{7}y$$

$$\text{So } \begin{pmatrix} -\frac{47}{7}y \\ y \\ 14y \end{pmatrix} \text{ e.g. } \begin{pmatrix} -\frac{47}{7} \\ 1 \\ 14 \end{pmatrix} \text{ or } \begin{pmatrix} -47 \\ 7 \\ 98 \end{pmatrix}$$

For $\lambda = 2$:

$$\begin{pmatrix} 0 & 5 & 3 \\ 0 & 7 & -1 \\ 0 & 0 & -7 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} 5y + 3z &= 0 \\ 7y - z &= 0 \\ -7z &= 0 \end{aligned}$$

So $z = 0$ and $y = 0$ any x

$$\text{So } \begin{pmatrix} x \\ 0 \\ 0 \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

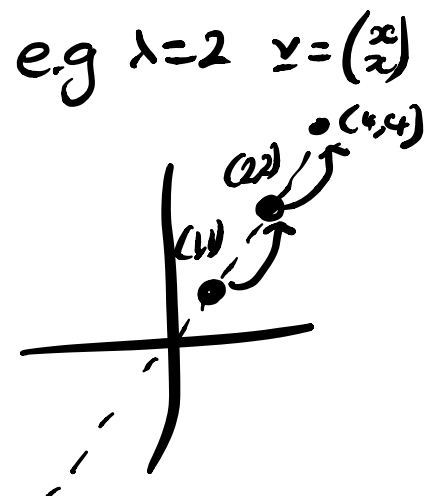
For $\lambda = 9$

$$\begin{pmatrix} -7 & 5 & 3 \\ 0 & 0 & -1 \\ 0 & 0 & -14 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} -7x + 5y + 3z &= 0 \\ -z &= 0 \\ -14z &= 0 \end{aligned}$$

$$\text{So } z = 0, y = \frac{7}{5}x \text{ so } \begin{pmatrix} x \\ \frac{7}{5}x \\ 0 \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ \frac{7}{5} \\ 0 \end{pmatrix} \text{ or } \begin{pmatrix} 5 \\ 7 \\ 0 \end{pmatrix}$$

Q7.

Eigenvalue, Eigenvalue
Pairs represent invariance
for which the points are
Scaled along the invariance
by the eigenvalue



Advanced skills

Q8. $M = \begin{pmatrix} 2 & 3 & 0 \\ 2 & 4 & 1 \\ 0 & -3 & 2 \end{pmatrix}$

a) $\begin{pmatrix} 2 & 3 & 0 \\ 2 & 4 & 1 \\ 0 & -3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$

$\Rightarrow 2 = \lambda, 0 = 0, -4 = -2\lambda$

So $\lambda = 2$ is an eigenvalue

b) $|M - \lambda I| = \begin{vmatrix} 2-\lambda & 3 & 0 \\ 2 & 4-\lambda & 1 \\ 0 & -3 & 2-\lambda \end{vmatrix} = \begin{vmatrix} 2-\lambda & 3 & 0 \\ 0 & 4-\lambda & 1 \\ -2(2-\lambda) & -3 & 2-\lambda \end{vmatrix}$

$\xrightarrow{C_1 \rightarrow C_1 - 2C_3}$

$= (2-\lambda) \begin{vmatrix} 1 & 3 & 0 \\ 0 & 4-\lambda & 1 \\ -2 & -3 & 2-\lambda \end{vmatrix}$

$\xleftarrow{\text{factorise}}$

$$= (2-\lambda) \begin{vmatrix} 1 & 3 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 3 & 2-\lambda \end{vmatrix} \quad \checkmark R_3 \rightarrow R_3 + 2R_1$$

$$= (2-\lambda) \begin{vmatrix} 4-\lambda & 1 \\ 3 & 2-\lambda \end{vmatrix} = (2-\lambda) ((4-\lambda)(2-\lambda)-3)$$

$$= (2-\lambda) (\lambda^2 - 6\lambda + 5)$$

$$= (2-\lambda) (\lambda-5) (\lambda-1) \quad \text{so } \lambda=1, 5 \text{ are others}$$

For $\lambda=1$:

$$\begin{pmatrix} 1 & 3 & 0 \\ 2 & 3 & 1 \\ 0 & -3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} x+3y &= 0 \\ 2x+3y+z &= 0 \\ -3y+z &= 0 \end{aligned}$$

$$\Rightarrow y = -\frac{1}{3}x, \quad z = 3y = -x \quad \text{so} \quad \begin{pmatrix} x \\ -\frac{1}{3}x \\ -x \end{pmatrix}$$

$$\text{e.g. } \begin{pmatrix} 1 \\ -\frac{1}{3} \\ -1 \end{pmatrix} \text{ or } \begin{pmatrix} 3 \\ -1 \\ -3 \end{pmatrix}$$

For $\lambda=5$:

$$\begin{pmatrix} -3 & 3 & 0 \\ 2 & -1 & 1 \\ 0 & -3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{aligned} -3x+3y &= 0 \\ 2x-y+z &= 0 \\ -3y-3z &= 0 \end{aligned}$$

$$\Rightarrow y = x, \quad z = -y = -x$$

$$\text{so } \begin{pmatrix} x \\ x \\ -x \end{pmatrix} \text{ e.g. } \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Q9. $\frac{dc}{dt} = 4c - 2f \quad \frac{df}{dt} = c + f$

$$a) \frac{d\underline{x}}{dt} = M \underline{x} \Rightarrow \begin{pmatrix} \frac{dc}{dt} \\ \frac{df}{dt} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} c \\ f \end{pmatrix}$$

$$\text{so } M = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$$

$$b) \underline{x} = e^{\lambda t} \underline{v} \Rightarrow \frac{d\underline{x}}{dt} = \lambda e^{\lambda t} \underline{v} = \lambda \underline{x}$$

$$\text{so } M \underline{x} = \lambda \underline{x}$$

$$c) |M - \lambda I| = 0 \Rightarrow (4 - \lambda)(1 - \lambda) + 2 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 6 = 0 \Rightarrow (\lambda - 3)(\lambda - 2) = 0$$

$$\Rightarrow \lambda = 2 \text{ or } 3$$

$$\text{For } \lambda = 2: \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = x \text{ so } \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda = 3$:

$$\begin{pmatrix} 1 & -2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow y = \frac{1}{2}x$$

so $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Then $\underline{x} = e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\underline{x} = e^{3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}$

both solve so does

$$\begin{aligned} \underline{x} &= A e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + B e^{3t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} A e^{2t} + 2B e^{3t} \\ A e^{2t} + B e^{3t} \end{pmatrix} \end{aligned}$$

so $c(t) = A e^{2t} + 2B e^{3t}$

$$f(t) = A e^{2t} + B e^{3t}$$